

His interpretation worked, provided that $E = hf$, where h was a new mathematical constant in nature, and is now known as Planck's constant in his honor.

It works like this. In any object, the energy is distributed among the atoms. A few have very little energy, a few have lots, while most have a middling amount of energy. But what we mean by "middling" changes as the temperature increases. Each atom can emit electromagnetic radiation in the form of quanta. For large values of f (high frequencies, or short wavelengths), the energy (E) needed to emit a single quantum is very large, and only a few atoms have that much energy.

At low frequencies (long wavelengths), it is easy to radiate the quanta because less energy is required. However, each quantum has so little energy (because f is small) that even added together, quanta do not contribute much to the spectrum. In the middle, though, there are lots of atoms producing lots of quanta, which add up to give the hump in the black body curve. Everything matched the experiments, for one particular value of h .

Planck did not think of these quanta as being like little particles, or bullets spat out by atoms. He thought there must be something about the internal workings of atoms that only allowed them to emit pulses of light, but that the light was still a wave. This is kind of like the way an ATM works. It will only let you have money in multiples of \$10, even though other amounts of money, such as \$27, exist.

Enter Einstein

Most of Planck's colleagues thought that his discovery, although useful, was only some kind of mathematical trick, and that when atoms were understood better, it would be possible to dispense with it. At first, the only

person who took and used Planck's new interpretation as it stood was Albert Einstein, then an unknown scientist just beginning his research. In 1905,

he showed that a perplexing phenomenon at the time, known as the photoelectric effect and which also had no classical explanation, could also be accounted for and understood if light did in fact travel as a stream of minute



a meeting of minds
Max Planck (left) and Albert Einstein were the two originators of quantum theory, although they came from different generations.

stepping up
Moving
around in
the quantum
world is like
moving up
a step-
ladder. You
can only
move by a
whole number of
steps; there is
nowhere "in
between" the steps
where you can rest
your feet.



"I spent 10 years of my life testing that 1905 equation of Einstein's, and contrary to my expectations, I was compelled in 1915 to assert its ambiguous verification in spite of its unreasonableness."

Robert Millikan (1868–1953)

particles. In its simplest form, the photoelectric effect is that a current can be started in a circuit by shining a light on a metal plate—electrons in the metal are agitated and create a current of electricity. The particular effects that Einstein

observed were only consistent with the current being generated by the electrons being struck by particles instead of by waves. However, at first, nobody else accepted the idea of these particles of light, and they were only given the name photons in 1926, after much more evidence emerged to show that Einstein was right.

light work
The photoelectric effect, explained in quantum terms by Albert Einstein in 1905, is the basis on which the solar panels on this futuristic electric vehicle convert sunlight into power.

light from the Sun generates an electric current in photoelectric cells

